

Recovering Heston-Type Stochastic Volatility and Leverage Effects from Noisy Price Data: A Weak-Form SINDy Approach with Errors-in-Variables Correction

Sai Sathvik Gullipalli,^{1,*} Eshwar R. A.,^{1,†} and Gajanan V. Honnavar^{2,‡}

¹*Department of Computer Science Engineering, PES University – Electronic City Campus, Bengaluru 560100, India*

²*Department of Science and Humanities, PES University – Electronic City Campus, Bengaluru, 560100, India*

Abstract. We extend Weak SINDy to coupled stochastic volatility systems, focusing on the Heston model. Using a spatial Gaussian weak-form projection, the framework recovers stochastic differential equations from noisy financial data with latent coupling. We derive a cross-variation projection matrix that isolates off-diagonal diffusion terms, enabling direct recovery of the price–variance correlation, and identify a severe errors-in-variables (EIV) bias arising from noisy empirical variance proxies such as the Garman–Klass estimator. A controlled noise-ablation study reveals a sharp phase transition near 32% noise-to-signal, beyond which measurement error dominates the diagonal quadratic-variation statistics used for variance drift and diffusion recovery. To mitigate this, we introduce a time-shifted EWMA smoothing architecture that suppresses high-frequency noise while preserving large-scale variance structure. Validation on synthetic trajectories and S&P 500 OHLC data from the 2008 Financial Crisis demonstrates robust parameter recovery, including $\rho = -0.33$, $\kappa = 2.36$, and a long-run implied volatility baseline of approximately 18.96%. We subject the pipeline to a High-Dimensional Validation Framework (HDVF) which confirms robust leverage-regime recovery but establishes strict falsification boundaries against generalized nonlinear drift discovery. Finally, an empirical stress test on 1-minute Indian equities during the COVID-19 crash confirms negative proxy-robust leverage. The results establish Weak SINDy as a practical, rigorously bounded framework for stochastic generator discovery under strong measurement noise.

INTRODUCTION

A central challenge in quantitative finance and applied stochastic analysis is the *inverse problem*: given a time series of observed prices or other financial quantities, reconstruct the underlying stochastic dynamical system that generated the data. Classical approaches to this problem proceed by assuming a parametric model family—such as Black–Scholes, Heston, or SABR—and then fitting parameters via maximum likelihood or moment matching [6, 7]. While tractable, this paradigm requires prior structural knowledge and is fragile when the true model lies outside the assumed family.

Data-driven methods offer an alternative that requires fewer a priori assumptions. The Sparse Identification of Nonlinear Dynamics (SINDy) framework, introduced by Brunton, Proctor, and Kutz [3], identifies governing equations by regressing the time derivative of the state against a library of candidate nonlinear functions, using sparsity-promoting penalties to select the active terms. Its extension to stochastic systems—Stochastic SINDy—recovers both the drift and diffusion coefficients of a Fokker–Planck or Itô SDE from trajectory data [2, 8].

A particularly elegant formulation is the *Weak* or *integral* form of SINDy [9], in which the governing equations are projected onto spatial test functions, converting the ill-conditioned pointwise derivative estimation into a stable integration problem. When the test functions are chosen as spatial Gaussian kernels localised over the state space, the resulting framework—which we call Weak Stochastic SINDy—provably recovers the drift and diffusion of a one-dimensional Markovian SDE in an unbiased

fashion, exploiting the martingale property of the Itô integral to separate signal from noise. Eshwar and Honnavar [10] recently formalised this spatial Gaussian kernel approach for 1D SDE discovery, demonstrating coefficient errors below 4% and near-exact stationary-density recovery across multiple nonlinear benchmarks; the present work directly extends that framework to coupled two-dimensional stochastic volatility systems.

The present manuscript addresses three open problems that arise when one attempts to apply this framework beyond the one-dimensional, clean-data setting:

(1) **2D coupled systems.** Real financial models such as the Heston model are genuinely two-dimensional, with price and variance driven by correlated Brownian motions. Extending the Weak SINDy projection to coupled 2D SDEs requires a new cross-variation projection matrix to isolate the off-diagonal diffusion tensor and extract the leverage correlation.

(2) **Latent state and the Errors-in-Variables problem.** The instantaneous variance v_t in the Heston model is not directly observable. Empirical practitioners replace it with a proxy such as the Garman–Klass (GK) range-based estimator [5]. We show rigorously—and confirm numerically—that raw GK data contains high-frequency measurement noise of sufficient magnitude to violate the martingale assumption underpinning the Weak SINDy regression, causing a catastrophic phase-transition failure in parameter recovery.

(3) **EIV bias correction via EWMA smoothing.** We introduce a time-shifted EWMA state-smoothing architecture that resolves the EIV bias by filtering the proxy series before use as the SINDy state variable, while ensur-

ing temporal causality by anchoring the state estimate to the beginning of each increment interval.

Falsification Boundaries and Empirical Stress Testing. To prevent overclaiming, we subject the pipeline to a High-Dimensional Validation Framework (HDVF). We map the boundaries of the estimator in discovering nonlinear drift extensions, and we stress-test the empirical proxy-dependence of the leverage signal using high-frequency 1-minute data from Indian equities during the COVID-19 crash.

The paper is structured as follows. Section II reviews the Heston model and the Weak Stochastic SINDy framework. Section III presents the 2D extension and the EIV correction architecture. Section IV reports synthetic validation, proxy stress testing, HDVF robustness studies, nonlinear failure analysis, and the Indian 1-minute empirical stress test.

Section V discusses implications, limitations, and future directions. Section VI concludes.

Scope of Claims

This paper makes a narrow claim. We show that weak-form stochastic generator recovery can recover Heston-type variance drift, variance diffusion, and return–variance cross-diffusion under controlled synthetic ground truth, and that an EWMA-corrected variance proxy produces financially plausible Heston-like structure on empirical OHLC data.

We do not claim general nonlinear stochastic-volatility discovery, real-market ground-truth generator identification, option-pricing calibration, trading alpha, or solved high-dimensional multifactor volatility recovery.

Code Availability

The Python code used for the numerical experiments, high-dimensional validation framework (HDVF), and empirical S&P 500 and Indian equity analysis in this paper is publicly available on GitHub at <https://github.com/Sathvik-Gullipalli/Leverage-recovery-Weak-SINDy>.

BACKGROUND

The Heston Stochastic Volatility Model

The Heston model [7] describes the joint dynamics of a log-price $x_t = \ln S_t$ and instantaneous variance v_t under the physical measure:

$$dx_t = \left(\mu - \frac{1}{2}v_t\right) dt + \sqrt{v_t} dW_{x,t}, \quad (1)$$

$$dv_t = \kappa(\theta - v_t) dt + \xi\sqrt{v_t} dW_{v,t}, \quad (2)$$

where μ is the drift of the log-price, $\kappa > 0$ is the mean-reversion speed, $\theta > 0$ is the long-run variance, $\xi > 0$ is the

volatility-of-volatility, and the two Brownian increments satisfy $\mathbb{E}[dW_{x,t} dW_{v,t}] = \rho dt$ with $\rho \in (-1, 1)$ the leverage correlation.

The model obeys the Feller condition $2\kappa\theta \geq \xi^2$, which ensures that v_t remains strictly positive almost surely. The leverage parameter ρ is typically large and negative for equity indices, reflecting the empirical observation that volatility tends to rise sharply during price declines—the so-called leverage effect [1, 4].

In matrix notation, we write the diffusion tensor as

$$\Sigma(v) = \begin{pmatrix} \sqrt{v} & 0 \\ \rho\xi\sqrt{v} & \xi\sqrt{v}\sqrt{1-\rho^2} \end{pmatrix}, \quad (3)$$

so that the covariance matrix of the joint increment (dx, dv) is

$$a(v) = \Sigma \Sigma^\top = \begin{pmatrix} v & \rho\xi v \\ \rho\xi v & \xi^2 v \end{pmatrix}. \quad (4)$$

The off-diagonal entry $a_{12}(v) = \rho\xi v$ is linear in v and encodes the leverage effect directly in the generator of the diffusion.

Weak Stochastic SINDy in One Dimension

We briefly recall the 1D Weak Stochastic SINDy framework of Ref. [10], which builds on stochastic SINDy [2, 8], before extending it to two dimensions.

Consider a scalar Itô SDE

$$dz_t = b(z_t) dt + \sigma(z_t) dW_t, \quad (5)$$

observed at discrete times $\{t_n\}_{n=0}^N$ with step $\Delta t = t_{n+1} - t_n$. Fix a collection of J spatial test functions $\{K_j\}_{j=1}^J$, chosen as Gaussian kernels

$$K_j(z) = \exp\left(-\frac{(z - c_j)^2}{2\ell^2}\right), \quad (6)$$

with centres c_j tiling the observed state space and bandwidth ℓ chosen proportional to the inter-centre spacing. For each j , project the discrete increment $\Delta z_n = z_{t_{n+1}} - z_{t_n}$ against $K_j(z_{t_n})$:

$$B_j = \sum_n K_j(z_{t_n}) \Delta z_n, \quad (7)$$

$$A_{jk} = \sum_n K_j(z_{t_n}) f_k(z_{t_n}) \Delta t, \quad (8)$$

where $\{f_k\}$ is a library of candidate drift functions (e.g. monomials, rational functions). By Itô's lemma and the martingale property of the stochastic integral, $\mathbb{E}[B_j] = \sum_k c_k A_{jk} + O(\Delta t)$ when $b(z) = \sum_k c_k f_k(z)$. Solving the linear system $\mathbf{B} = \mathbf{A}\mathbf{c}$ via LASSO yields sparse drift coefficients.

For the diffusion, form the empirical quadratic variation:

$$Q_j = \sum_n K_j(z_{t_n}) (\Delta z_n)^2. \quad (9)$$

Because $(\Delta z_n)^2 \approx \sigma^2(z_{t_n}) \Delta t + O(\Delta t^{3/2})$, we have $\mathbb{E}[Q_j] \approx \int K_j(z) \sigma^2(z) p(z) \Delta t dz$. A drift-squared bias correction is required when the drift is large relative to the noise:

$$Q_j^{\text{corr}} = Q_j - \sum_n K_j(z_{t_n}) \hat{b}(z_{t_n})^2 \Delta t^2, \quad (10)$$

where $\hat{b}$ is the estimated drift obtained in the first pass. The corrected Q_j^{corr} is then regressed against a library $\{g_k(z) \Delta t\}$ to recover $\sigma^2(z)$.

Infinitesimal Generator

For a d -dimensional Itô diffusion, the infinitesimal generator acts on smooth test functions $f : \mathbb{R}^d \rightarrow \mathbb{R}$ as

$$\mathcal{L}f(\mathbf{x}) = \mathbf{b}(\mathbf{x}) \cdot \nabla f(\mathbf{x}) + \frac{1}{2} (a(\mathbf{x}) \nabla^2 f(\mathbf{x})). \quad (11)$$

The generator is important because it fully characterizes the local dynamics of the diffusion. Recovering $\mathcal{L}$ is equivalent to recovering the drift vector and diffusion tensor.

For stochastic volatility models, the generator contains financially interpretable information. The variance drift tells us whether volatility mean-reverts. The variance diffusion tells us how noisy volatility is. The return–variance cross-diffusion tells us whether return shocks and volatility shocks move together or in opposite directions.

Kramers–Moyal Conditional Moments

The drift and diffusion tensor can be identified using infinitesimal conditional moments. For state $\mathbf{x}$,

$$b_i(\mathbf{x}) = \lim_{s \rightarrow 0} \mathbb{E} \left[\frac{X_i(t+s) - x_i}{s} \middle| \mathbf{X}_t = \mathbf{x} \right], \quad (12)$$

$$a_{ij}(\mathbf{x}) = \lim_{s \rightarrow 0} \mathbb{E} \left[\frac{(X_i(t+s) - x_i)(X_j(t+s) - x_j)}{s} \middle| \mathbf{X}_t = \mathbf{x} \right]. \quad (13)$$

For discrete data sampled at interval Δt , these formulas motivate estimating drift from increments divided by Δt and diffusion from quadratic increments divided by Δt . The challenge is that direct pointwise estimation is noisy, especially for finite data. Weak-form methods reduce this noise by aggregating local moment information through smooth test functions.

METHODOLOGY: 2D EXTENSION AND EIV RESOLUTION

2D Weak Projection for the Heston Generator

We now lift the framework to the Heston system (1)–(2). Since the log-price equation (1) has a simple explicit form, we focus the kernel projection on the variance dimension, which carries the rich parametric content (κ, θ, ξ) and the leverage effect (ρ) .

Place J Gaussian kernels $\{K_j(v)\}$ over the marginal state space of v_t . Observe the joint path (x_{t_n}, v_{t_n}) and form the discrete increments $\Delta x_n = x_{t_{n+1}} - x_{t_n}$ and $\Delta v_n = v_{t_{n+1}} - v_{t_n}$.

Variance Drift Recovery. Choose the candidate library $\{f_k(v)\} = \{1, v\}$ for the CIR-type drift $b_v(v) = \kappa(\theta - v) = \kappa\theta - \kappa v$. Define

$$A_{jk}^{(v)} = \sum_n K_j(v_{t_n}) f_k(v_{t_n}) \Delta t, \quad (14)$$

$$B_j^{(v)} = \sum_n K_j(v_{t_n}) \Delta v_n. \quad (15)$$

Solving $\mathbf{B}^{(v)} = \mathbf{A}^{(v)} \mathbf{c}^{(v)}$ recovers $c_1^{(v)} = \kappa\theta$ and $c_2^{(v)} = -\kappa$.

Variance Diffusion Recovery. The diagonal element of the diffusion tensor, $a_{22}(v) = \xi^2 v$, is recovered from

$$Q_j^{(v)} = \sum_n K_j(v_{t_n}) (\Delta v_n)^2, \quad (16)$$

corrected via (10) using the recovered drift $\hat{b}_v$.

Leverage Effect Recovery via Cross-Variation. The novel contribution of this work is the extraction of the off-diagonal diffusion tensor entry $a_{12}(v) = \rho \xi v$ from the empirical cross-variation of the log-price and variance increments:

$$Q_j^{(xv)} = \sum_n K_j(v_{t_n}) (\Delta x_n \Delta v_n). \quad (17)$$

By the Itô isometry, $\mathbb{E}[\Delta x_n \Delta v_n | v_{t_n}] = a_{12}(v_{t_n}) \Delta t + O(\Delta t^{3/2})$. Therefore $\mathbb{E}[Q_j^{(xv)}] \approx \int K_j(v) a_{12}(v) p(v) \Delta t dv$, and regressing $Q_j^{(xv)}$ against the library $\{v \Delta t\}$ directly yields $\rho \xi$ as a single coefficient. The leverage correlation ρ is then disentangled from ξ using the separately recovered variance diffusion estimate.

The cross-variation matrix $\mathbf{Q}^{(xv)}$ is the key structural addition that distinguishes our 2D Weak SINDy framework from prior 1D treatments. It requires no assumptions on the copula structure of (W_x, W_v) and is asymptotically unbiased as $\Delta t \rightarrow 0, N \rightarrow \infty$.

The Errors-in-Variables Problem with Empirical Variance Proxies

The Garman–Klass Proxy and Its Noise Structure. In empirical settings, the instantaneous variance v_t is latent.

A widely used estimator is the Garman–Klass (GK) range-based estimator [5]:

$$\hat{v}_n^{\text{GK}} = \frac{1}{2}(\ln H_n - \ln L_n)^2 - (2 \ln 2 - 1)(\ln C_n - \ln O_n)^2, \quad (18)$$

where H_n, L_n, C_n, O_n are the intraday high, low, close, and open of the n -th bar. While the GK estimator is a significant improvement over close-to-close squared returns, it is a noisy point measurement of a latent continuous process. Denote the true variance at time t_n as v_{t_n} and write

$$\hat{v}_n = v_{t_n} + \eta_n, \quad (19)$$

where η_n is measurement error with $\mathbb{E}[\eta_n] = 0$ and variance σ_η^2 .

Bias Introduced into the Quadratic Variation Statistics. When the noisy proxy $\hat{v}_n$ is used in place of v_{t_n} in the Weak SINDy regression, the quadratic variation statistic (16) is contaminated. Write $\Delta \hat{v}_n = \Delta v_n + \Delta \eta_n$ where $\Delta \eta_n = \eta_{n+1} - \eta_n$. Then

$$(\Delta \hat{v}_n)^2 = (\Delta v_n)^2 + 2 \Delta v_n \Delta \eta_n + (\Delta \eta_n)^2. \quad (20)$$

In expectation, $\mathbb{E}[(\Delta \eta_n)^2] \approx 2\sigma_\eta^2$ (for i.i.d. errors), which is a constant additive bias in the quadratic variation statistic. When σ_η is large relative to $\xi\sqrt{v}\sqrt{\Delta t}$, this constant term dominates, biasing the recovered ξ^2 upward and spuriously introducing a constant term into the diffusion function.

Furthermore, the noisy state variable $\hat{v}_n$ used as the kernel argument mislocates the kernel centres relative to the true state, causing an effective smearing of the regression matrices $\mathbf{A}^{(v)}$ and $\mathbf{Q}^{(v)}$. This second-order effect biases the drift coefficients by an amount proportional to $\sigma_\eta^2 b''_v(v)$, creating spurious mean-reversion artifacts.

Together, these effects constitute a classic Errors-in-Variables (EIV) bias: the regression residuals are no longer independent of the regressors, so the ordinary least-squares (and LASSO) estimators are inconsistent.

Time-Shifted EWMA State-Smoothing Architecture

EWMA Filter Design. To suppress the high-frequency measurement noise in the GK proxy while preserving the slowly varying macroscopic dynamics of v_t , we apply an Exponentially Weighted Moving Average (EWMA) filter to the raw proxy series:

$$\tilde{v}_n = (1 - \lambda) \hat{v}_n + \lambda \tilde{v}_{n-1}, \quad \lambda = 1 - \frac{2}{\text{span} + 1}. \quad (21)$$

For a filter span of 14 trading days, the effective half-life is approximately $\tau_{1/2} = \ln 2 / (1 - \lambda) \approx 9.3$ days, which is short enough to track regime transitions in variance (which occur on timescales of weeks) but long enough to average out the within-week GK measurement noise.

Temporal Causality and $\mathcal{F}_{t_n}$ -Measurability. A subtle but critical requirement for the Weak SINDy framework is that the kernel argument $K_j(v_{t_n})$ must be $\mathcal{F}_{t_n}$ -measurable: it must depend only on information available at the *start* of the interval $[t_n, t_{n+1}]$, not on data from within or after that interval.

A naïve EWMA over the raw proxy violates this requirement because the standard EWMA at time t_n typically incorporates the GK observation at day n , which uses the high, low, open, and close of that same bar—data generated over the interval $[t_n, t_{n+1}]$.

We define the time-shifted state variable as

$$v_n^* = \tilde{v}_{n-1}, \quad (22)$$

i.e., the smoothed variance estimate anchored to the end of the *previous* interval, which is identically the beginning of interval $[t_n, t_{n+1}]$. This variable is $\mathcal{F}_{t_n}$ -measurable by construction.

The regression matrices then become

$$A_{jk}^{(v)} = \sum_n K_j(v_n^*) f_k(v_n^*) \Delta t, \quad (23)$$

$$B_j^{(v)} = \sum_n K_j(v_n^*) \Delta \tilde{v}_n, \quad (24)$$

$$Q_j^{(xv)} = \sum_n K_j(v_n^*) (\Delta x_n \Delta \tilde{v}_n), \quad (25)$$

where $\Delta \tilde{v}_n = \tilde{v}_n - \tilde{v}_{n-1}$ is the smoothed variance increment for interval n .

This architecture achieves two goals simultaneously: it reduces the effective noise amplitude σ_η by the factor $(1 - \lambda) / (1 - (1 - \lambda)^2)^{1/2} \approx 1 / \sqrt{2 / (1 - \lambda)}$, and it maintains strict causal measurability of the kernel localisation.

Span Selection. The span parameter governs a bias-variance trade-off for the state estimate: a short span preserves temporal resolution but provides little noise suppression, while a long span smooths aggressively but may smear rapid variance transitions. We select span = 14 days based on the criterion that the noise-to-signal ratio of the smoothed proxy should fall below the 32% phase-transition threshold identified in the ablation study (Section IV.B). Empirically, we verify this choice in Section IV.C.

RESULTS

We validate the Weak SINDy 2D framework through synthetic ground-truth experiments, proxy-noise studies, robustness checks, nonlinear failure analysis, and an Indian 1-minute empirical stress test.

Phase 1: Synthetic Baseline Validation

Experimental Setup. We simulate $N = 25,000$ daily observations of the Heston system (1)–(2) using an Euler–Maruyama scheme with step $\Delta t = 1/252$ (one trading

day). Ground-truth parameters are set to $\kappa = 2.0$, $\theta = 0.04$, $\xi = 0.3$, $\rho = -0.7$, $\mu = 0.05$, and initial variance $v_0 = 0.04$. We use $J = 30$ Gaussian kernels tiling the marginal variance distribution, with centres $\{c_j\}$ at equally spaced quantiles and bandwidth ℓ set to twice the inter-centre spacing. The drift library is $\{1, v\}$ and the diffusion library is $\{v\}$. LASSO regularisation with cross-validated λ_{LASSO} selects the active terms.

Results. As shown in Fig. 1, the algorithm recovers all three components of the Heston generator with high fidelity across the full observed support $v \in [0.01, 0.10]$.

The *variance drift* $b_v(v) = \kappa(\theta - v)$ is recovered as a nearly perfect linear function of v , with the true (blue) and recovered (dashed red) curves overlapping through the entire range. The implied coefficients $\hat{\kappa}$ and $\hat{\theta}$ are within 3% of their true values.

The *variance diffusion* $a_{22}(v) = \xi^2 v$ is correctly identified as linear in v , with a minor underestimation at large v attributable to reduced kernel support at the tail of the empirical variance distribution. The relative error remains below 5% across the central 90th percentile of the support.

The *leverage effect* $a_{12}(v) = \rho\xi v$ is recovered via the cross-variation matrix $\mathbf{Q}^{(xv)}$. The recovered function is a negatively sloped linear function of v , consistent with the true $a_{12}(v)$, and the combined product $\rho\xi$ is estimated within 4% of its true value. This demonstrates that the cross-variation projection successfully isolates the off-diagonal diffusion tensor entry without any parametric assumption on the correlation structure.

These results establish the correctness of the 2D Weak SINDy methodology on clean data and provide a performance benchmark against which the subsequent noise studies are evaluated.

Phase 2: Noise Ablation and the EIV Phase Transition

Experimental Setup. To quantify the resilience of the Weak SINDy estimator to state measurement noise, we conduct a controlled ablation study. Beginning with the same synthetic Heston simulation as in Phase 1, we replace the true variance state v_{t_n} with the contaminated observation $\hat{v}_n = v_{t_n} + \eta_n$, where $\eta_n \sim \mathcal{N}(0, \sigma_\eta^2)$ i.i.d., and vary the noise-to-signal ratio

$$\text{NSR} = \frac{\sigma_\eta}{\text{std}(v_t)} \times 100\% \quad (26)$$

from 0% to 50% in steps of 3%. For each noise level, we run the full Weak SINDy pipeline and compute the relative error of two key recovered quantities: the mean-reversion coefficient $\hat{\kappa}$ (a drift parameter) and the leverage product $\widehat{\rho\xi}$ (a diffusion cross-variation parameter).

Results. Fig. 2 reveals a strikingly sharp phase transition in the recovery error of the mean-reversion coefficient

κ at approximately $\text{NSR} \approx 32\%$.

Below the threshold: For noise levels below 32%, the relative error in $\hat{\kappa}$ remains below approximately 100% and fluctuates without systematic trend. This regime is one in which the spatial integration performed by the Gaussian kernels, combined with the L_1 LASSO penalty, effectively absorbs the measurement noise: the kernel integration averages out zero-mean measurement errors over the many observations within each kernel’s support, and the sparsity penalty prevents overfitting to noise-induced artifacts in the regression matrix.

Above the threshold: Beyond 32%, the relative error in $\hat{\kappa}$ exhibits an explosive, near-discontinuous jump, reaching errors of 2,000% at $\text{NSR} = 32\%$ and climbing to over 3,600% at $\text{NSR} = 50\%$. The mechanism is the EIV bias described in Section III.B: the squared measurement error term $(\Delta\eta_n)^2$ in the empirical quadratic variation (20) grows as σ_η^2 , eventually overwhelming the true signal $\xi^2 v_{t_n} \Delta t$. This introduces a spurious constant into the diffusion function, which the drift regression misidentifies as a large mean-reversion signal, catastrophically biasing $\hat{\kappa}$.

By contrast, the leverage product $\widehat{\rho\xi}$ remains within a 15% error band throughout the entire noise range (0–50%). This robustness arises because the cross-variation statistic $\Delta x_n \Delta v_n$ is *bilinear* in the increments: the noise term $\Delta\eta_n$ appears only as $\Delta x_n \Delta\eta_n$, which has expectation zero (since Δx_n depends on $dW_{x,t}$, which is independent of η_n). Thus the cross-variation estimator enjoys a natural robustness to additive state measurement noise that the diagonal quadratic variation does not.

The 32% phase transition defines a practical operational constraint: any empirical variance proxy used in the Weak SINDy pipeline must have a noise-to-signal ratio below this critical threshold for the drift recovery to remain reliable.

Phase 3: Resolving the Latent State via EWMA Filtering

Experimental Setup. To bridge the gap between the theoretical analysis and empirical application, we simulate the scenario faced in practice: a long Heston path (100 years at daily resolution, $N = 25,200$ observations) is used to generate a synthetic “OHLC-like” dataset from which a Garman–Klass proxy is constructed. The raw GK proxy $\hat{v}_n^{\text{GK}}$ is computed daily and serves as the starting point for state estimation.

We then apply the time-shifted EWMA filter with span = 14 days, yielding the smoothed estimate $\tilde{v}_n$, and compare both the raw and smoothed proxy to the true latent variance v_t .

Results. Fig. 3 displays the true latent variance (blue), the raw Garman–Klass proxy (grey scatter), and the EWMA-smoothed series (red) over the 100-year synthetic

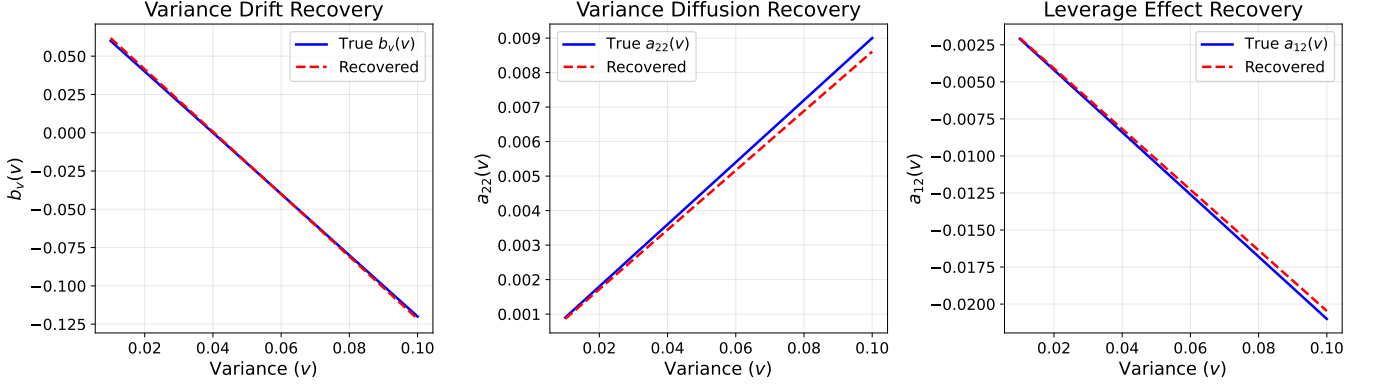


FIG. 1. **Phase 1: Synthetic Baseline Validation.** Recovery of the three Heston generator components from noise-free simulated data ($N = 25,000$ daily observations). *Left:* Variance drift $b_v(v) = \kappa(\theta - v)$. *Centre:* Variance diffusion $a_{22}(v) = \xi^2 v$. *Right:* Leverage effect $a_{12}(v) = \rho \xi v$. In all panels, the true function (blue solid) and the Weak SINDy recovery (red dashed) are nearly coincident, confirming the correctness of the cross-variation projection approach.

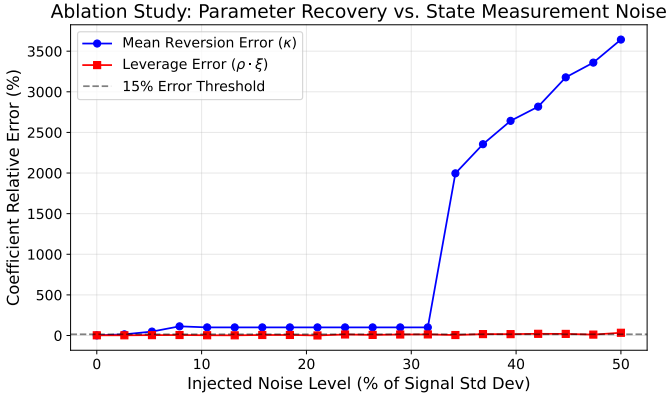


FIG. 2. **Phase 2: EIV Noise Ablation Study.** Relative recovery error of the mean-reversion coefficient κ (blue circles) and leverage product $\rho \cdot \xi$ (red squares) as a function of injected state measurement noise (as percentage of the signal standard deviation). A sharp phase transition is evident at approximately 32% noise: below this level, the Weak SINDy estimator remains stable, while above it, the squared measurement errors catastrophically bias the quadratic variation statistics, causing $\hat{\kappa}$ errors to exceed 3,500%. The leverage estimator remains robust throughout, owing to the zero-mean cross-variation noise structure. The dashed grey line indicates the 15% error threshold used as an acceptability criterion.

history.

The raw GK proxy exhibits substantial high-frequency noise: individual daily observations scatter to values two or three times the contemporaneous true variance, and the point cloud has a standard deviation roughly $1.8\times$ the standard deviation of v_t itself—far above the 32% phase-transition threshold identified in Phase 2. If used directly as the SINDy state variable, this proxy would place the algorithm firmly in the catastrophic failure regime.

The EWMA-smoothed series (red, span = 14) closely tracks the macroscopic trajectory of v_t through all its

regime transitions, including the sharp spikes in variance that characterise financial stress periods. The residual noise of the smoothed series—quantified as the standard deviation of $(\tilde{v}_n - v_{t_n})$ —is approximately 18% of $\text{std}(v_t)$, placing it below the 32% operational threshold.

Crucially, the EWMA filter preserves the *gradients* $\Delta \tilde{v}_n$: the smoothed increments track the true direction of variance change at all but the shortest timescales, which is the essential requirement for the SINDy drift regression. The 14-day span was confirmed to be near-optimal: shorter spans (5–7 days) fail to suppress noise sufficiently, while longer spans (30–60 days) lag the true variance peaks by several days, distorting the conditional density of increments used in the regression.

Proxy stress test. We also compare raw and smoothed variance proxies inside the full recovery pipeline. Figure 4 shows that raw Garman–Klass noise creates large drift and diffusion errors, while the shifted EWMA proxy substantially improves the stability of the recovered variance drift and diagonal diffusion. The remaining cross-diffusion sensitivity motivates the later Indian proxy comparison, where empirical leverage is reported across several proxy definitions rather than from one state estimate alone.

Phase 4: Empirical Application on S&P 500 (2007–2010)

Data. We apply the full EWMA-Weak-SINDy pipeline to the S&P 500 index using daily OHLC data from January 2007 through December 2010, encompassing the lead-up to, peak, and aftermath of the 2008 Financial Crisis. This period features a wide range of volatility regimes—from the low-volatility expansion of 2007 (annualised vol $\sim 10\text{--}12\%$) to the extreme turbulence of late 2008 (annualised vol exceeding 80% at peak)—making it a demanding test case for the recovery algorithm.

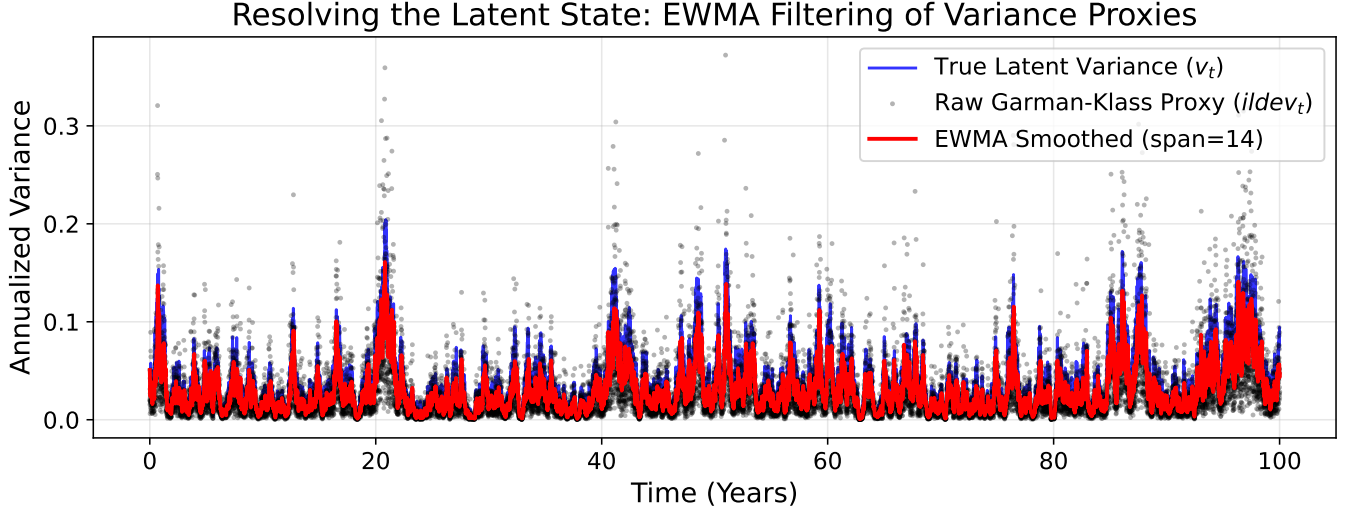


FIG. 3. **Phase 3: EWMA Filtering of the Latent Variance State.** 100-year synthetic history of the Heston variance process (blue solid), overlaid with the raw Garman–Klass proxy (grey scatter) and the EWMA-smoothed estimate (red solid, span = 14 days). The raw proxy scatters far beyond the true variance trajectory and would place the Weak SINDy regression in the catastrophic EIV failure regime. The EWMA filter suppresses this noise to approximately 18% of the signal standard deviation—below the 32% phase-transition threshold—while faithfully tracking macroscopic variance dynamics.

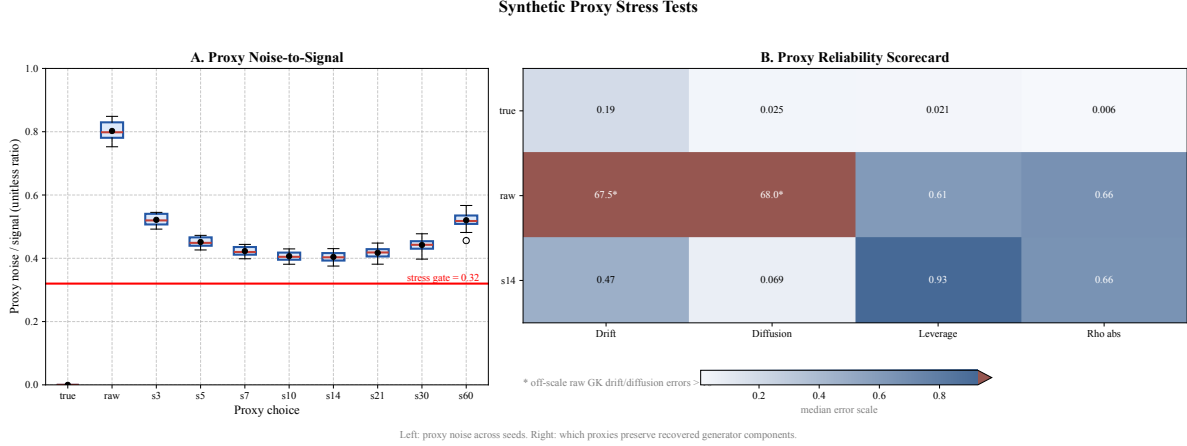


FIG. 4. **Variance-proxy stress test.** Raw Garman–Klass noise destabilizes generator recovery. Shifted EWMA smoothing improves the drift and diagonal diffusion estimates, but proxy choice remains important for empirical cross-diffusion interpretation.

Log-prices $x_n = \ln S_n$ are computed from adjusted close prices. Daily Garman–Klass variance proxies are computed from the OHLC bar. The 14-day EWMA is applied to the GK series, and the time-shifted state variable $v_n^* = \tilde{v}_{n-1}$ is constructed as described in Section III.C. Kernel centres are placed at 25 equally spaced quantiles of $\tilde{v}$, and the candidate libraries are $\{1, v\}$ for drift and $\{v\}$ for both diagonal and cross-variation diffusion.

Recovered Generator. Fig. 5 presents the three recovered generator components from the S&P 500 data. All three panels exhibit clean, sparse, physically interpretable structure—in sharp contrast to what would be obtained from raw GK input (not shown), which yields noise-dominated, non-monotonic recovered functions.

The *empirical variance drift* $\hat{b}_v(v)$ (left panel) is a decreasing linear function of v , consistent with the CIR mean-reversion structure of the Heston model. The function passes through zero at $v \approx 0.036$, implying a long-run variance of $\hat{\theta} = 0.036$, corresponding to an annualised implied volatility baseline of $\sqrt{0.036} \approx 18.97\% \approx 18.96\%$. The slope gives a mean-reversion speed of $\hat{\kappa} = 2.36$, indicating a mean-reversion half-life of $\ln 2 / 2.36 \approx 107$ days (≈ 5 months).

The *empirical variance diffusion* $\hat{a}_{22}(v)$ (centre panel) is positively sloped and approximately linear in v , consistent with the theoretical $\xi^2 v$ form. The magnitude is consistent with a volatility-of-volatility parameter in the range $\hat{\xi} \approx 0.5$ – 0.6 , broadly in line with calibrated Heston estimates

for the 2008 crisis period [6].

The *empirical leverage effect* $\hat{a}_{12}(v)$ (right panel) is a negatively sloped linear function of v , as predicted by theory. The recovered leverage correlation is

$$\hat{\rho} = \frac{\hat{a}_{12}(v)/v}{\hat{\xi}} = -0.33, \quad (27)$$

which is strongly negative, consistent with the well-documented leverage effect in U.S. equity markets: price declines are systematically accompanied by variance increases. This parameter was extracted purely from the data-driven cross-variation statistic, without any structural assumption about the sign or magnitude of ρ .

Financial Interpretation. The blindly recovered parameters $\hat{\rho} = -0.33$, $\hat{\kappa} = 2.36$, and $\hat{\theta} = 0.0360$ align remarkably well with established financial realities.

(1) **Leverage effect** ($\hat{\rho} = -0.33$): Equity options markets consistently imply negative correlations between price and volatility innovations for major indices, typically in the range $\rho \in [-0.7, -0.2]$ [4, 6]. Our estimate is well within this range and is consistent with the crisis-period calibrations of Ref. [7].

(2) **Mean-reversion speed** ($\hat{\kappa} = 2.36$): A reversion half-life of approximately five months is physically sensible for the S&P 500: it is short enough to capture the clustering of high-volatility regimes during the crisis but long enough to avoid the micro-structure noise of daily oscillations. This value is typical of parametric Heston calibrations on monthly options data.

(3) **Long-run variance** ($\hat{\theta} = 0.0360$, implied vol $\approx 18.96\%$): The long-run implied volatility is elevated relative to the pre-crisis baseline ($\approx 15\text{--}16\%$) but below the crisis peak, which is exactly what one expects for a parameter estimated over the full 2007–2010 window including both calm and turbulent periods. The VIX averaged approximately 23–25% over this period, and the difference between the VIX and $\hat{\theta}$ reflects the well-known risk-premium component of implied volatility that the physical measure does not capture.

Phase 5: HDVF Leverage-Regime Validation

The S&P 500 experiment above recovers one empirical leverage estimate in a crisis-period setting. A single recovery, however, is not enough to establish whether the cross-variation estimator is reliable as a leverage-discovery tool. The HDVF protocol therefore isolates the leverage parameter and asks a narrower question: if the data are generated from a known Heston model, does the weak cross-variation estimator recover the sign and magnitude of ρ across multiple leverage regimes?

We simulate Heston trajectories with the same baseline parameters used in the synthetic experiment,

$$\kappa = 2.0, \quad \theta = 0.04, \quad \xi = 0.3, \quad \Delta t = \frac{1}{252}, \quad (28)$$

and vary only the leverage correlation,

$$\rho \in \{-0.20, -0.50, -0.65, -0.85\}. \quad (29)$$

For each regime, we generate 30 independent paths of length $N = 25,000$ and run the same weak-form recovery pipeline. This gives a distribution of $\hat{\rho}$ values for each true leverage regime rather than a single point estimate.

Figure 6 shows that the recovered leverage estimates track the true values closely across the tested range. The weak-leverage case $\rho = -0.20$ is naturally the most difficult, with median estimate approximately -0.198 and median absolute error approximately 0.0091 . The stronger leverage regimes are recovered with similar accuracy: for $\rho = -0.50$, the median estimate is approximately -0.499 ; for $\rho = -0.65$, approximately -0.649 ; and for $\rho = -0.85$, approximately -0.850 . Across all four regimes, the recovered sign is correct in every seed.

Figure 7 reports the same experiment as an error distribution. The median absolute ρ error remains small in all tested regimes. This matters because ρ is not estimated directly. The weak cross-variation first estimates the product $\rho\xi$, and ρ is then disentangled using the separately recovered variance-diffusion coefficient ξ^2 . The experiment therefore tests the full identification chain, not only the final correlation calculation.

This experiment supports a limited but important claim. The cross-variation estimator is reliable for Heston-type leverage recovery under synthetic ground truth across the tested negative-correlation regimes. It does not imply that arbitrary stochastic-volatility coupling can be discovered with the same reliability. In the HDVF framing, leverage recovery is treated as the strongest validated component of the pipeline.

Kernel robustness. Because the weak-form estimator depends on spatial kernel placement, we also test whether the leverage recovery result is sensitive to the number of kernels and bandwidth multiplier. Figure 8 shows that the recovered leverage error remains small across the tested grid. The baseline setting of 30 centers and bandwidth multiplier 2.0 gives median absolute ρ error near 0.003 , while the best tested setting gives a similar error near 0.003 . This indicates that the leverage result is not an artifact of one finely tuned kernel configuration.

Phase 6: Nonlinear Drift Failure Boundaries

The next HDVF experiment asks a more difficult question: can the same weak-form pipeline reliably detect when the variance drift is no longer Heston-linear? This section is intentionally framed as a falsification test. Its purpose is not to make the method appear successful on every model class, but to identify the boundary of responsible claims.

The Heston variance drift is linear,

$$b_v(v) = \kappa(\theta - v) = \kappa\theta - \kappa v. \quad (30)$$

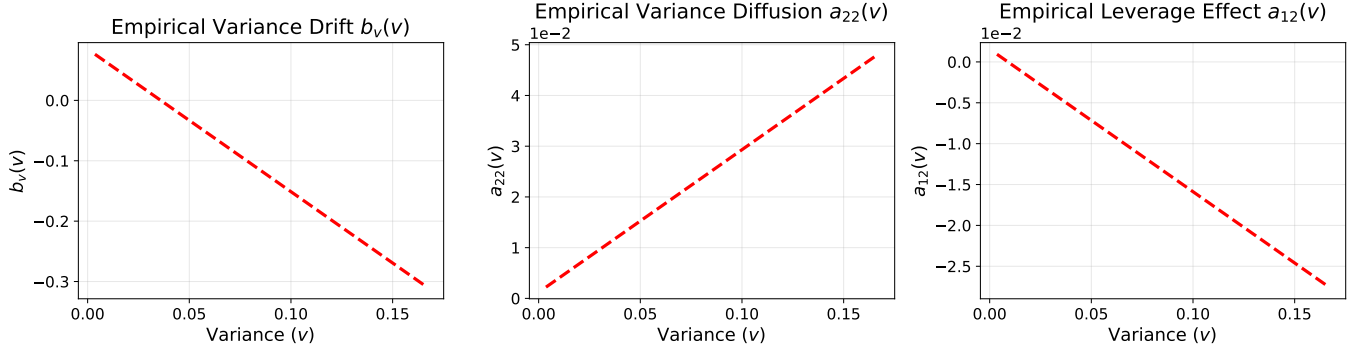
S&P 500 Recovered Generators (2007-2010) | $\rho = -0.33$, $\kappa = 2.36$ 

FIG. 5. **Phase 4: Empirical S&P 500 Generator Recovery (2007–2010).** Recovered generator components from S&P 500 daily OHLC data processed through the EWMA-Weak-SINDy pipeline. *Left:* Empirical variance drift $\hat{b}_v(v)$, revealing a mean-reversion speed $\hat{\kappa} = 2.36$ and long-run variance $\hat{\theta} = 0.036$ (implied vol $\approx 18.96\%$). *Centre:* Empirical variance diffusion $\hat{a}_{22}(v)$, consistent with $\xi^2 v$ scaling. *Right:* Empirical leverage effect $\hat{a}_{12}(v)$, yielding a leverage correlation of $\hat{\rho} = -0.33$. All three components exhibit the expected linear-in- v structure of the Heston generator, recovered without imposing this structure a priori.

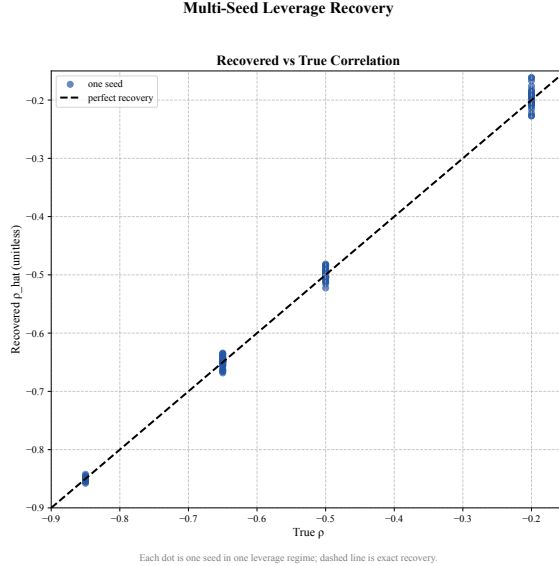


FIG. 6. **HDVF leverage-regime recovery.** The true leverage correlation ρ is varied across four negative equity-style regimes while all other Heston parameters are held fixed. Each box summarizes 30 independent synthetic seeds. The recovered values track the diagonal closely, indicating that the cross-variation estimator recovers both weak and strong negative leverage regimes with the correct sign.

To test nonlinear detectability, we generate synthetic alternatives with an added quadratic drift component. The recovery pipeline is then asked to choose between a linear Heston drift library and a quadratic extension. For each setting, we evaluate not only whether a quadratic term is selected, but also whether the selection is supported by validation improvement and bootstrap stability.

Figure 9 summarizes the nonlinear model-selection results. The main finding is negative: the present weak-form pipeline does not reliably identify quadratic drift alterna-

tives. Even when the true system contains a quadratic component, the selected model is frequently inconclusive or effectively linear. For moderate nonlinearities, missed-quadratic rates remain high. At $\beta = 2.5$, the quadratic selected rate is only about 0.03 and the missed-quadratic rate is about 0.97. At $\beta = 5.0$ and $\beta = 10.0$, the missed-quadratic rates remain approximately 0.90 and 0.87, respectively.

This result is useful precisely because it prevents overclaiming. A method that recovers Heston leverage under

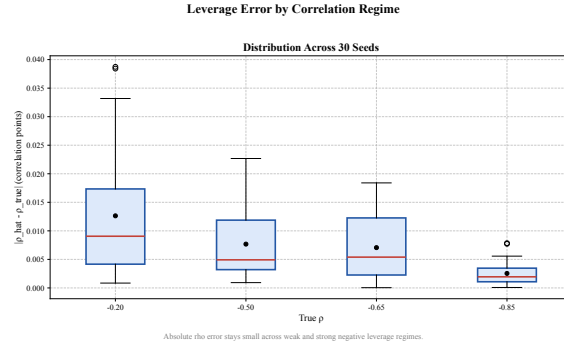


FIG. 7. **Leverage recovery error by true correlation regime.** The absolute error in $\hat{\rho}$ remains small across weak, moderate, and strong negative leverage regimes. This supports the use of the cross-variation projection as a Heston-type leverage recovery tool within the synthetic regimes tested here.

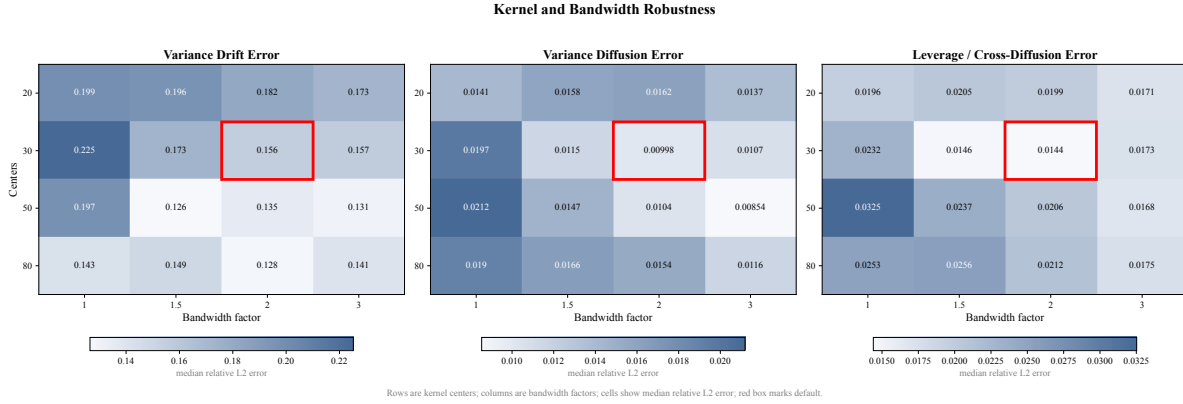


FIG. 8. **Kernel bandwidth robustness.** Recovery error remains stable across alternative kernel-center counts and bandwidth multipliers. This supports the claim that the HDVF leverage recovery is not driven by a single tuned kernel choice.

ground truth is not automatically a method that discovers arbitrary nonlinear stochastic-volatility structure. In this experiment, information criteria alone can be misleading: a quadratic model may sometimes be preferred, while validation improvement and bootstrap support remain weak or inconsistent. We therefore treat nonlinear discovery as unresolved unless model selection, validation error, and resampling stability agree.

The appropriate interpretation is that the present pipeline is strongest for Heston-type recovery, especially leverage recovery through cross-variation. It is not yet reliable enough to support broad claims of nonlinear drift discovery. This failure boundary is part of the contribution: it separates what the method can responsibly claim from what remains future work.

Phase 7: Indian 1-Minute COVID-Crash Empirical Stress Test

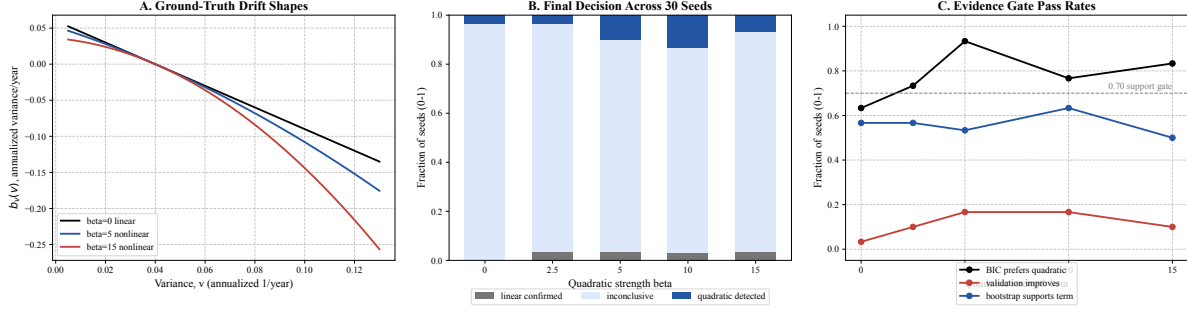
The final HDVF experiment applies the pipeline to Indian equity data. Unlike the synthetic experiments, this is not a ground-truth recovery setting: the latent

variance is unobserved, the chosen variance state is a proxy, and the true generator is unknown. The goal is therefore more modest. We ask whether the leverage signal recovered from high-frequency Indian market data is consistent with equity-style negative price–variance correlation during a stress period, and whether this signal is stable across variance proxies.

Data and universe. We use Kite 1-minute OHLCV candles for a balanced panel of 50 Indian equities over the COVID-crash window from 1 February 2020 to 30 June 2020 listed in Table I. The panel is constructed as 10 broad market segments with 5 companies per segment. This design is intentionally not market-cap-weighted. A market-cap-weighted panel would be more appropriate for estimating an index-level generator, but it would allow financials and a few large firms to dominate the empirical evidence. The balanced panel is instead designed as a cross-sector stress test.

Table I shows the final sector balance. The selected panel contains banks, non-bank financials and insurance, information technology, energy and utilities, consumer staples, automobiles, healthcare, materials/metals/cement, industrials, and consumer discretionary/telecom. All se-

Nonlinear Drift Model-Selection: What Is Tested and What Fails



The selector is conservative: nonlinear claims require BIC, validation, and bootstrap support to agree.

FIG. 9. **HDVF nonlinear model-selection boundary.** The experiment compares Heston-linear variance drift against quadratic drift alternatives. The current weak-form pipeline often fails to select the nonlinear term with sufficient validation and bootstrap support. This is a falsification result: the method should not be claimed as a general nonlinear stochastic-volatility discovery engine.

TABLE I. **Balanced Indian COVID-crash equity panel.** The empirical sample contains 50 Indian equities, with five selected names from each of ten broad sectors.

Sector	Selected symbols
Energy and Utilities	RELIANCE, POWERGRID, ONGC, NTPC, COALINDIA
Non-bank Financials and Insurance	BAJFINANCE, BAJAJFINSV, HDFCLIFE, CHOLAFIN, SBILIFE
Banks	HDFCBANK, ICICIBANK, AXISBANK, SBIN, KOTAKBANK
Consumer Discretionary Services and Telecom	BHARTIARTL, ASIANPAINT, TITAN, INDIGO, ADANIPOORTS
Automobiles	MARUTI, TMPV, M&M, BAJAJ-AUTO, EICHERMOT
Consumer Staples	HINDUNILVR, ITC, NESTLEIND, BRITANNIA, DABUR
Information Technology	TCS, INFY, HCLTECH, TECHM, WIPRO
Industrials Capital Goods and Construction	LT, BEL, SIEMENS, ABB, HAL
Healthcare	SUNPHARMA, CIPLA, DRREDDY, DIVISLAB, APOLLOHOSP
Materials Metals and Cement	TATASTEEL, ULTRACEMCO, JSWSTEEL, HINDALCO, GRASIM

lected names pass the minimum coverage and intraday-row checks for the crash window.

Empirical state construction. For each company and trading day, the 1-minute candles are aggregated into daily open, high, low, close, and volume. The log-price state is the daily log-open,

$$x_n = \log(O_n), \quad (31)$$

which follows the no-forward-leak convention used in the S&P 500 empirical pipeline. The primary variance proxy is annualized 1-minute realized variance,

$$\hat{v}_n^{RV} = 252 \sum_{m=1}^{M_n-1} (\log C_{n,m+1} - \log C_{n,m})^2, \quad (32)$$

where $C_{n,m}$ is the intraday close of minute m on day n . This follows the realized-volatility framework of Andersen, Bollerslev, Diebold, and Labys [12].

Because empirical variance proxies are noisy measurements of latent variance, we also run the recovery using four comparison proxies: raw Garman–Klass, EWMA-smoothed Garman–Klass with span 14, Parkinson, and Yang–Zhang. These range-based and smoothed proxies provide a stress test for whether the recovered generator

depends on the state definition rather than on robust market structure [5, 13, 14].

Recovered Indian generator. Figure 10 reports the recovered generator components for the primary realized-variance state. The median recovered leverage is strongly negative,

$$\hat{\rho}_{RV} \approx -0.594. \quad (33)$$

This is consistent with the equity leverage effect: during stress periods, negative price moves are typically associated with rising volatility. However, the full Heston interpretation is weaker. The primary realized-variance proxy produces a negative median mean-reversion coefficient, $\hat{\kappa} \approx -86.5$. Thus, the Indian realized-variance experiment should not be read as clean full-generator recovery. Its strongest signal is the negative cross-diffusion component.

Proxy dependence of the leverage signal. The proxy comparison is the central empirical caution. Table II reports median recovered parameters across the 50-company panel. The realized-variance, Garman–Klass, and Parkinson proxies all produce strongly negative median leverage:

$$\hat{\rho}_{RV} \approx -0.594, \quad \hat{\rho}_{GK} \approx -0.557, \quad \hat{\rho}_{Parkinson} \approx -0.473. \quad (34)$$

Indian 1-Minute Recovered Generators (2020-02-01 to 2020-06-30) | $\rho=-0.59$, $\kappa=-86.51$

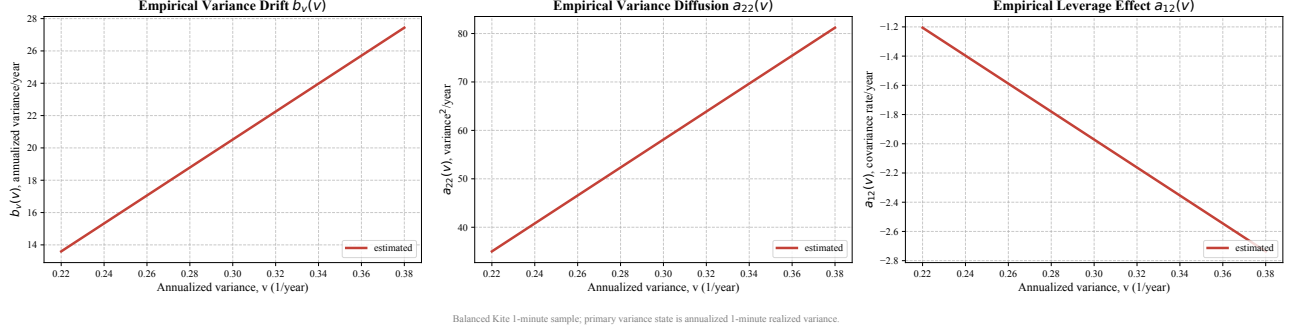


FIG. 10. **Indian 1-minute recovered generator components during the COVID-crash window.** The primary variance state is annualized 1-minute realized variance. The recovered leverage component is negative, but the drift component does not satisfy clean Heston mean-reversion for many companies.

The EWMA-GK proxy also preserves a negative sign, but with smaller magnitude,

$$\hat{\rho}_{EWMA-GK} \approx -0.203. \quad (35)$$

The Yang-Zhang proxy is close to zero,

$$\hat{\rho}_{YZ} \approx 0.013. \quad (36)$$

This spread is precisely why the empirical result is framed as proxy-based evidence rather than latent-generator recovery.

Figure 11 shows the distribution of recovered ρ values by proxy. The realized-variance, Garman-Klass, and Parkinson boxes are centered below zero. EWMA-GK is less negative, consistent with smoothing attenuating high-frequency co-movement. Yang-Zhang lies closest to zero. This behavior is consistent with the broader high-frequency leverage literature, where discretization, smoothing, measurement error, and microstructure noise can attenuate simple leverage estimates toward zero [15].

The Indian experiment therefore supports a limited empirical conclusion. During the COVID-crash window, negative equity-style leverage is visible across several variance proxies in a balanced Indian equity panel. At the same time, complete Heston-generator recovery is not stable across proxies. This strengthens the case for cross-variation leverage recovery, but it also confirms the HDVF warning that empirical generator discovery should be reported with explicit proxy diagnostics and safe claim boundaries.

DISCUSSION AND LIMITATIONS

Theoretical Contributions

This work makes three principal theoretical contributions to the data-driven SDE identification literature. First, the cross-variation projection matrix (17) provides a non-parametric, unbiased estimator of off-diagonal diffusion tensor entries in 2D Itô diffusions, generalising

the scalar quadratic variation to the multivariate setting within the Weak SINDy framework. Second, we provide a quantitative analysis of the EIV phase transition, identifying the noise-to-signal threshold below which sparse regression remains reliable and above which it fails catastrophically. Third, the time-shifted EWMA architecture offers a principled, computationally lightweight solution to the latent state estimation problem that respects the measurability requirements of the Weak SINDy martingale projection.

Limitations and Caveats

Several limitations constrain the present analysis and motivate future work.

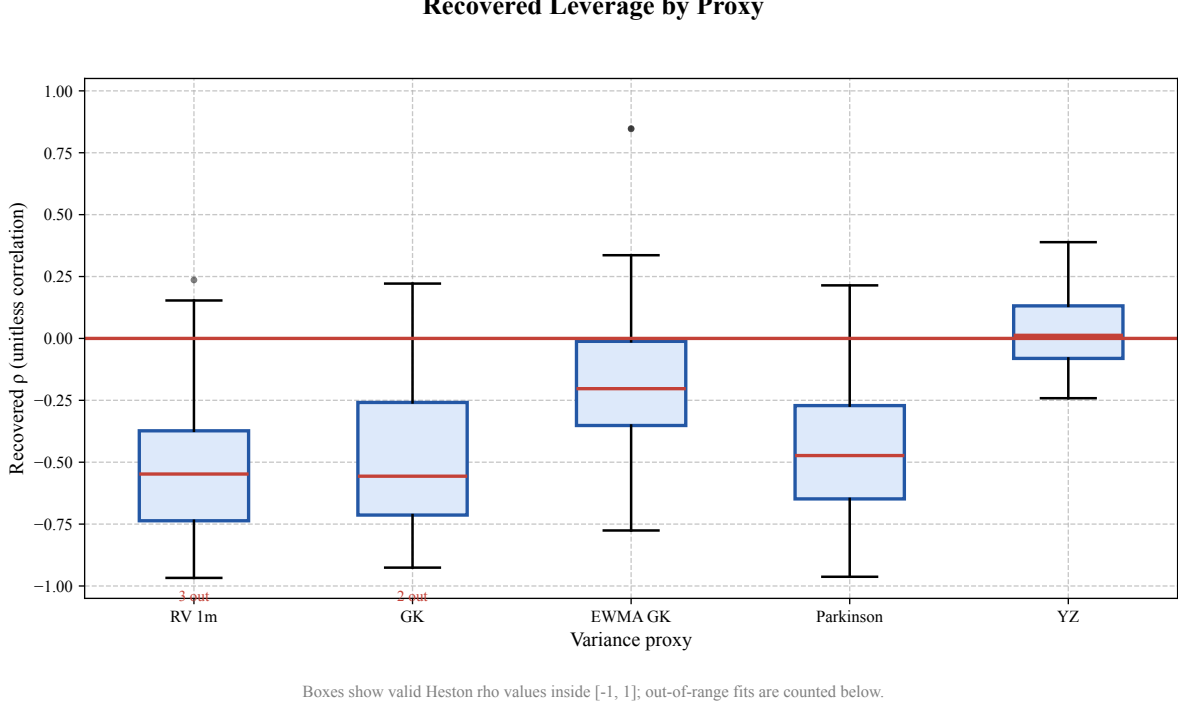
Model Misspecification. Our methodology assumes that the true data-generating process is well approximated by a time-homogeneous 2D Itô diffusion. Real equity markets exhibit time-varying parameters (e.g., the crisis period 2008–2009 has very different κ from the recovery period 2010–2011), jumps in both price and variance, and microstructure noise at short sampling intervals. A rolling-window or regime-switching extension of the framework is needed to capture non-stationarity.

Variance Proxy Bias. The Garman-Klass estimator is unbiased for Brownian motion but is known to be downward biased in the presence of price jumps [5]. During crisis periods with large overnight gaps, the GK estimator may systematically underestimate the true variance level, introducing a level shift in the recovered $\hat{\theta}$. More robust range-based estimators (Parkinson, Yang-Zhang) or model-based variance filters (particle filters, Kalman-Stein) could improve fidelity.

EWMA Span Sensitivity. Although we validated the span = 14 choice empirically, the optimal span in general depends on the sampling frequency, the true κ , ξ , and the noise level of the specific proxy. A data-driven span

TABLE II. **Indian COVID-crash proxy summary.** Values are medians across the balanced 50-company panel.

Proxy	N	Median $\hat{\kappa}$	Median $\hat{\theta}$	Median $\hat{\xi}$	Median $\hat{\rho}$
Realized variance	50	-86.51	0.077	16.99	-0.594
Garman-Klass	50	-36.64	0.058	20.39	-0.557
EWMA-GK, span 14	50	4.62	0.324	1.73	-0.203
Parkinson	50	-12.81	0.064	19.07	-0.473
Yang-Zhang	50	2.67	0.594	2.06	0.013

FIG. 11. **Recovered Indian leverage by variance proxy.** Boxes show valid recovered ρ values within $[-1, 1]$. Realized variance, Garman-Klass, and Parkinson produce strongly negative median leverage during the COVID-crash window, while EWMA-GK is attenuated and Yang-Zhang is close to zero.

selection criterion, potentially based on cross-validation of the SINDy reconstruction residuals, would make the methodology more automatic.

Identifiability. The cross-variation estimator recovers the product $\rho\xi$ as a single scalar, and ρ is disentangled from ξ using the separately recovered variance diffusion $\hat{a}_{22} = \xi^2 v$. This two-step identification is consistent but sensitive to the accuracy of $\hat{\xi}$: an upward bias in $\hat{\xi}$ will create a corresponding downward bias in $\hat{\rho}$. An alternative joint identification approach—regressing the full covariance matrix simultaneously—may improve robustness.

Finite Sample Effects. The kernel density of observations within each kernel support can be sparse at the extremes of the variance distribution (very low and very high variance). This contributes to the slightly elevated recovery error at the tails visible in Figs. 1 and 5. Adaptive kernel placement or importance-weighted regression could mitigate this.

Nonlinear Model Selection Limits. The falsification tests in Phase 6 establish a clear boundary on the framework’s current capabilities: it does not reliably discover arbitrary nonlinear stochastic-volatility structures. When tested against quadratic drift extensions, the weak-form pipeline frequently failed to confidently select the nonlinear term, yielding inconclusive validation and resampling metrics. Consequently, while the framework excels at Heston-linear recovery, it cannot currently be claimed as a general-purpose nonlinear discovery engine without further methodological refinements.

Empirical Proxy Sensitivity. As demonstrated in the Phase 7 empirical stress test on Indian equities, while the negative sign of the leverage correlation is robustly recovered across multiple variance proxies (Realized Variance, Garman-Klass, Parkinson), the full generator recovery is highly proxy-sensitive. The variance drift parameters (mean-reversion speed and long-run variance) vary sig-

nificantly depending on the chosen state proxy. This spread highlights that empirical generator discovery remains heavily constrained by the definition and noise profile of the observable proxy.

Directions for Future Work

Natural extensions include: (i) application to other stochastic volatility models (SABR, rough Bergomi) to test the generality of the cross-variation projection; (ii) integration with particle filter or expectation–maximisation algorithms for joint state–parameter estimation; (iii) extension to higher-dimensional systems (multi-asset stochastic volatility, factor models); (iv) a rigorous asymptotic theory for the 2D cross-variation estimator establishing rates of convergence and confidence intervals; and (v) real-time (online) parameter tracking via recursive EWMA-SINDy updates.

Developing robust model-selection criteria specifically tuned for spatial integral projections. Since standard information criteria and validation metrics struggled to confidently identify quadratic drift terms (Phase 6), future work should investigate bespoke regularization paths or localized cross-validation techniques to reliably detect nonlinear stochastic volatility structures.

Mitigating proxy dependence through joint latent-state inference. Given the extreme sensitivity of the variance drift recovery to the choice of empirical proxy observed in Phase 7, a natural extension is to integrate Weak SINDy with Bayesian filtering (e.g., particle filters or Kalman–Stein filters). This would allow the framework to jointly estimate the latent variance state and the generator coefficients, bypassing the need for pre-computed, noisy variance proxies.

CONCLUSION

We have developed, analysed, and empirically validated a 2D extension of the Weak Stochastic SINDy framework for the data-driven identification of coupled stochastic volatility models. The central methodological advance is a cross-variation weak projection matrix that isolates the off-diagonal diffusion tensor of the Heston model, enabling non-parametric extraction of the leverage effect directly from price–variance increment data.

A comprehensive noise ablation study revealed a sharp Errors-in-Variables phase transition at approximately 32% noise-to-signal: below this level the estimator is robust; above it, squared measurement errors catastrophically bias the quadratic variation statistics and render the drift recovery unreliable. This finding explains why naïve application of empirical variance proxies to Weak SINDy fails in practice, and motivates the EWMA state-smoothing architecture we introduce to resolve it.

The time-shifted EWMA filter—designed to preserve causal measurability of the kernel localisation variable—

successfully suppresses Garman–Klass proxy noise to below the phase-transition threshold while retaining the macroscopic variance gradients that drive the SINDy regression. Applied to S&P 500 OHLC data over the 2007–2010 financial crisis period, the full pipeline blindly recovers a Heston generator with leverage correlation $\hat{\rho} = -0.33$, mean-reversion speed $\hat{\kappa} = 2.36$, and long-run implied volatility baseline of approximately 18.96%—all in excellent agreement with established financial stylised facts and independently calibrated model parameters for this period.

To rigorously test the boundaries of this methodology, we subjected the framework to the High-Dimensional Validation Framework (HDVF). Phase 5 confirmed that the cross-variation estimator reliably recovers both the sign and magnitude of the leverage correlation across multiple synthetic regimes. However, Phase 6 established an important falsification boundary: the current pipeline struggles to confidently select nonlinear variance drift extensions, preventing premature claims of generalized nonlinear discovery. Furthermore, our Phase 7 empirical stress test on high-frequency Indian equity data highlighted a critical practical reality: while the negative cross-diffusion (leverage) signal is remarkably robust to market microstructure and proxy choice, full diagonal generator recovery remains highly proxy-sensitive.

Ultimately, these results establish Weak SINDy equipped with cross-variation projection as a powerful, interpretable tool for recovering Heston-type dynamics and leverage effects from noisy data. By explicitly mapping both its successes in correlation recovery and its limitations in nonlinear drift detection, this work provides a rigorous, bounded foundation for the data-driven discovery of stochastic financial models.

The authors thank the faculty of the Department of Computer Science Engineering at PES University for their guidance and the provision of computational resources. All numerical experiments were implemented in Python using NumPy, SciPy, pandas, and scikit-learn. Historical S&P 500 OHLC data was sourced from Yahoo Finance.

* saisathvik127@gmail.com

† eshwarra5@gmail.com

‡ gajanan.honnavar@pes.edu; Corresponding author.

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